



MCS Thesis Defence

Comparative Benchmarking of CatBoost, TabPFN and TabFM for Explainable Multi-Tier Hair Fall Risk Stratification

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Outline

- 1 Background & problem
- 2 Research questions & objectives
- 3 Literature gap
- 4 Methodology & data
- 5 Results
- 6 Explainability (SHAP)
- 7 Practical costs
- 8 Conclusion & future work

Why hair fall risk?

Many small factors add up — no single test predicts it



Nutrition

iron • protein • vitamin D • calcium



Stress & sleep

stress level • late sleep • disturbance



Health & heredity

liver enzyme ALT • anaemia • family history



Lifestyle

chemical treatments • hard water

Today

- Self-reported history, late clinical check
- Subjective and hard to access
- Data are tabular — ideal for machine learning

**Goal: an accurate and explainable risk tier —
Low, Moderate or High**

Problem statement

Three gaps in existing work



Conflicting evidence

Ensembles best in one study,
Random Forest beats CatBoost in
another — no clear model choice.



Foundation models untested

TabPFN and TabFM predict without
training, but were never tested on
hair fall data.



Black-box predictions

Accurate models are hard to explain
— clinicians cannot check the
reason.

This study: a fair benchmark of CatBoost, TabPFN and TabFM on the same data — with SHAP for every model.

Research questions → objectives

General objective: benchmark CatBoost, TabPFN and TabFM for explainable hair fall risk stratification

	Question	Specific objective
1	Performance Can zero-shot models match a tuned CatBoost?	Compare accuracy, macro-F1, ROC-AUC + McNemar test
2	Data efficiency How does training size change results?	Retrain with 100, 500, 2,000 rows
3	Explainability Which features drive predictions?	SHAP for all three models
4	Practicality What do the models cost to run?	Runtime and implementation issues

What the literature says

32 sources reviewed



Tuned trees

Gradient boosting (XGBoost, LightGBM, CatBoost) is still the strong baseline on tabular data.



Tabular foundation models

TabPFN (2023, 2025) and TabFM (2026) predict in one forward pass — no tuning.



SHAP

Shapley values explain any model, so different models can be compared.



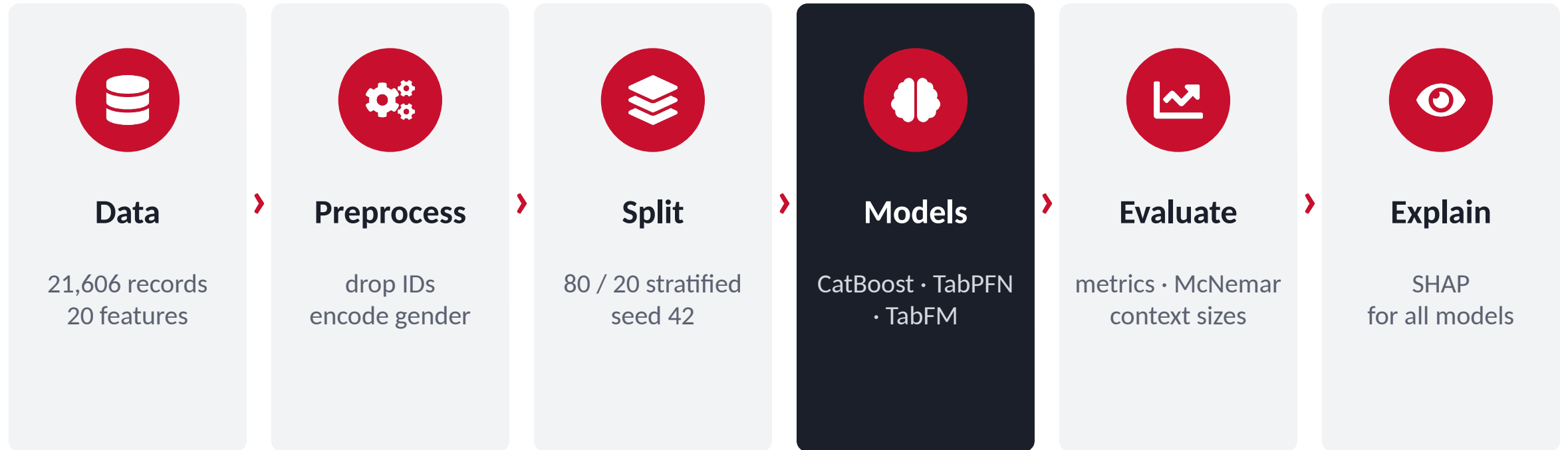
Hair loss studies

Classical models, conflicting results, few explanations.

Gap: no study compares two tabular foundation models with a tuned CatBoost on hair fall data, with SHAP for all.

Methodology

Quantitative, experimental benchmark — same data, split and metrics for every model



Tools: Python · scikit-learn · catboost · tabpfn · tabfm · shap · statsmodels | Kaggle & Google Colab, NVIDIA T4 GPU

Dataset

Assembled from two public datasets (Kaggle, Mendeley) and adjusted with domain knowledge

21,606

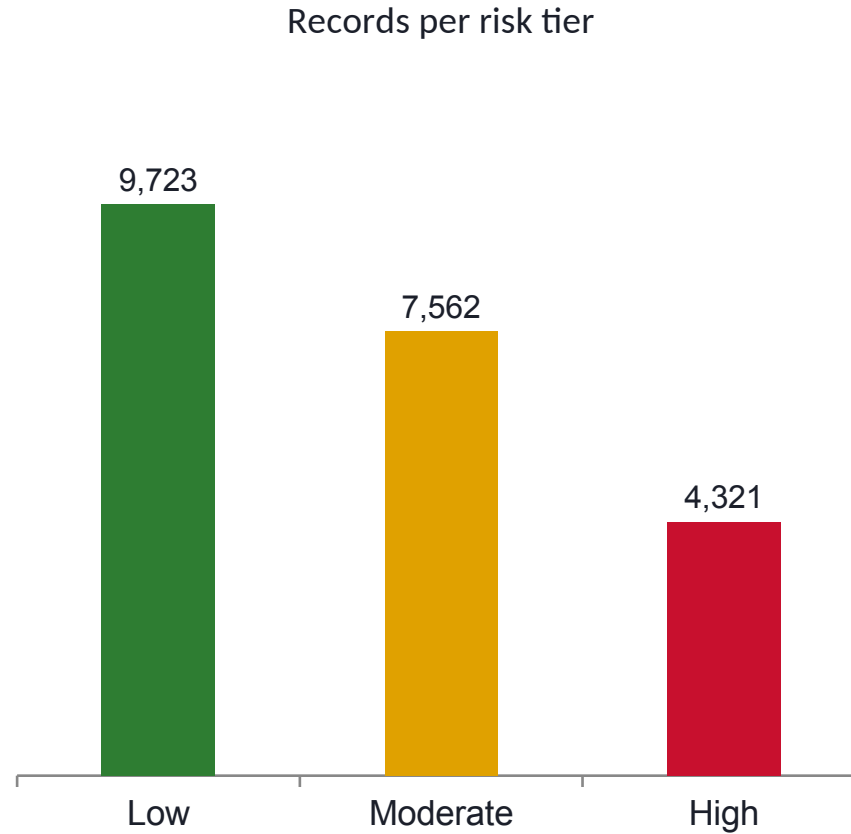
records

20

predictors

3

risk tiers



Biomarkers

iron, protein, vitamin D, calcium, manganese, ALT, body water, stress score

Clinical & lifestyle flags

anaemia, chronic illness, stress, sleep, chemicals, hard water

Demographics

age, gender, family history

Three models, two philosophies

Tuned training vs. zero-shot prediction

CatBoost

Gradient-boosted trees

- Grid search: depth $\times$ learning rate
- 5-fold stratified CV
- Best: depth 4, lr 0.1
- 17,284 training rows

TabPFN

Pretrained transformer

- No training, no tuning
- Training rows given as context
- One forward pass
- 17,284 context rows

TabFM

Google, 2026 • row + column attention

- No training, no tuning
- Single ensemble member
- Slow at large context
- 500 context rows

Result 1 — performance is almost identical

Held-out test set: 4,322 records

 CatBoost

78.51%

Macro-F1 0.776 · AUC 0.919

 TabPFN

78.57%

Macro-F1 0.779 · AUC 0.923

 TabFM

77.88%

Macro-F1 0.772 · AUC 0.918

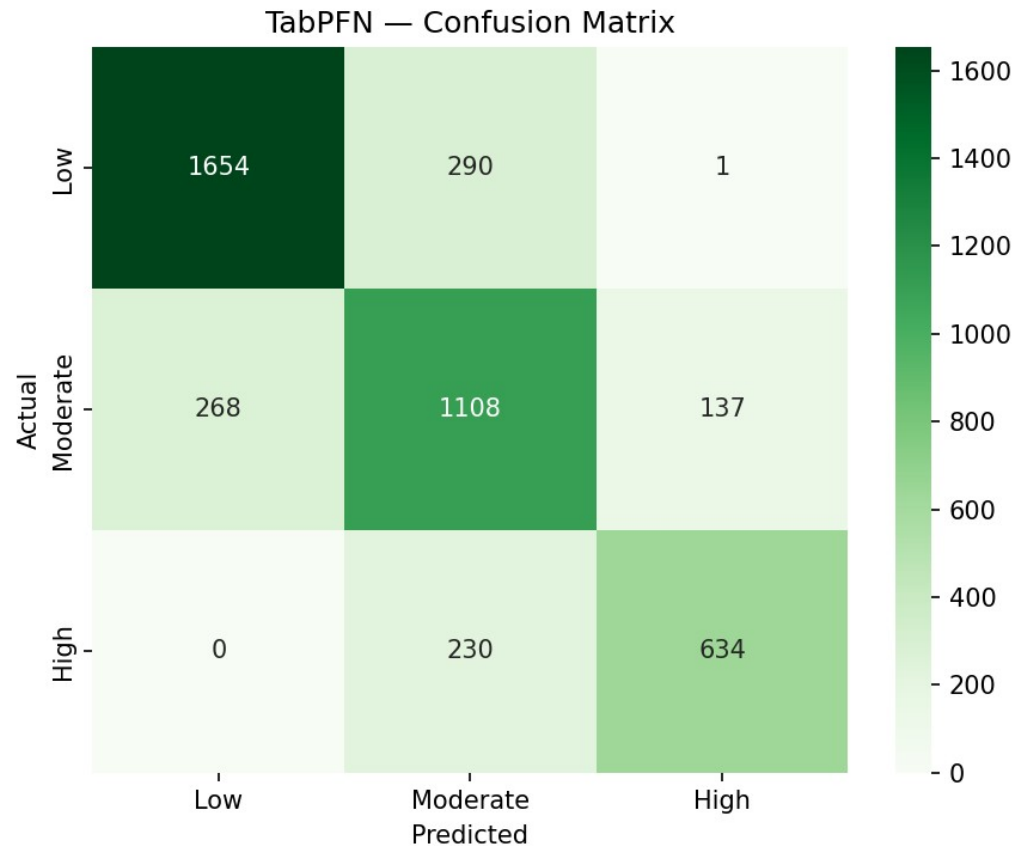


McNemar test: no significant difference

$p = 0.90 \cdot 0.20 \cdot 0.11$ — TabPFN matched tuned CatBoost with zero tuning; TabFM used < 3% of the data.

Where the models make mistakes

Errors stay between neighbouring tiers



Low ↔ High almost never

only 1–3 of ~930 errors



Moderate is hardest

F1 = 0.71 for every model



High often under-rated

27–31% of High predicted as Moderate



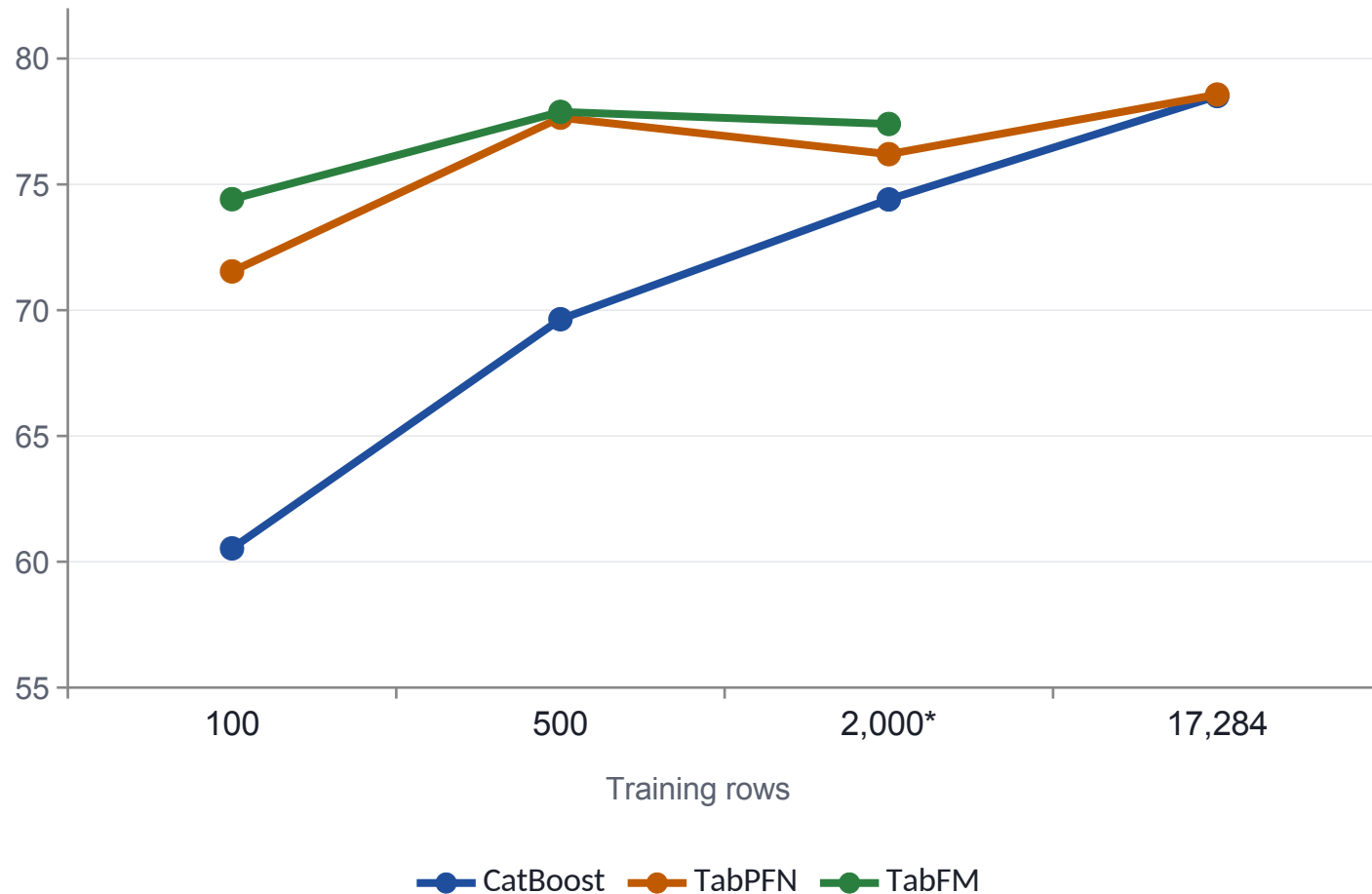
Ordinal agreement

Quadratic weighted kappa ≈ 0.81

TabPFN confusion matrix (others look the same)

Result 2 — foundation models need far less data

Accuracy (%) when models get only part of the training data



* 2,000-row run tested on a 500-record subset. TabFM not run at full size.

+11 to +14

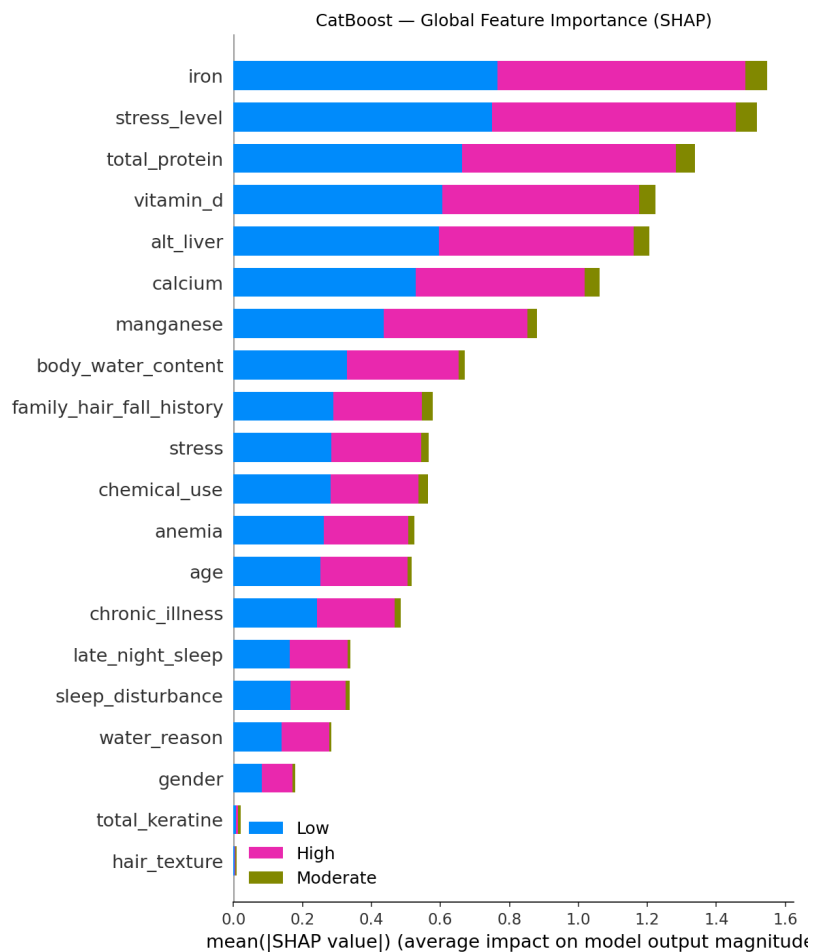
points over CatBoost with only 100 rows

≥ 5× less data

to pass 70% accuracy; the gap closes at full data

Result 3 — the same features drive every model

SHAP explanations agree across all three models



CatBoost global SHAP importance

Top 7 in all three models

- Iron
- Stress level
- Total protein
- Vitamin D
- ALT (liver)
- Calcium
- Manganese

▲ Raise risk

high stress level, high ALT, every risk flag

▼ Lower risk

high iron, protein, vitamin D, calcium, manganese

Result 4 — no tuning ≠ cheap to run

TabFM prediction time grows steeply with context size

0.11 s

per record

500-row context

1.8 s

per record

2,000-row context

did not finish

17,284-row context

4× more context → ~16× slower per record

Choice depends on data size and number of predictions: CatBoost is cheap to run once trained; foundation models need no training but pay at every prediction.

Software problems solved



SHAP Permutation → numba error; used KernelExplainer



TabFM crashed on 1-row input; padded to 2 rows



TabPFN checkpoint load; patched torch.load

Conclusion



RQ1 • Performance

No significant difference: tuned CatBoost $\approx$ zero-shot TabPFN $\approx$ TabFM (~78%).



RQ2 • Data efficiency

Foundation models far better with little data; gap closes at full size.



RQ3 • Explainability

Same 7 biomarkers and directions in all models — SHAP works for all.



RQ4 • Practicality

TabFM is slow at large context; new tools need testing before use.

When labelled data is scarce, start with a tabular foundation model — and always explain the predictions.

Limitations & future work



Limitations

- Constructed dataset — not clinical validation
- TabFM used 500 rows, not full data
- SHAP for foundation models on 20–30 records
- One split, one seed; early stopping used test set



Future work

- Validate on real clinical data
- Full-context TabFM, several seeds and splits
- Larger SHAP samples
- Ordinal models for the Moderate tier
- Ensembles and lightweight distilled models

Thank you

Questions & discussion



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